Is Appearance Free Action Recognition Possible?

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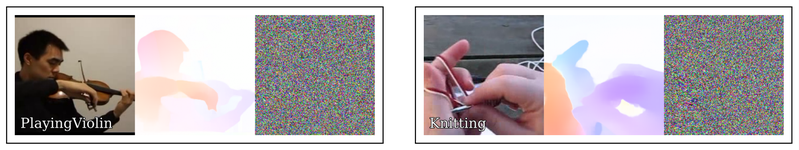
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Appearance biases in Action Recognition

Static information is often enough to correctly guess the action



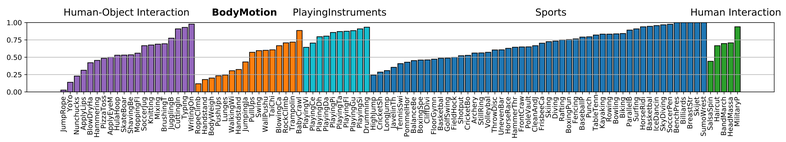
Our contribution consist of:

- showing that current SoTA are heavily biased towards appearance
- introducing a syntethic dataset with no discriminatory static cues
- highlighting the importance of explicit motion modeling in networks

Why are static biases a problem?

Single frame performance is just too good
This suggests that contemporary deep-learning architectures for video understanding can perform well on the task of action recognition with little to no regard for the actual scene dynamics.

Top-1 Accuracy on single frame UCF101 by *Action Groups*



- Top-1 average on single UCF101 frames 65.6%
- Top-1 average on single UCF101 optical flow frames 29.4%

This is way too good considering no dynamic information is present!

Scientifically unsatisfying

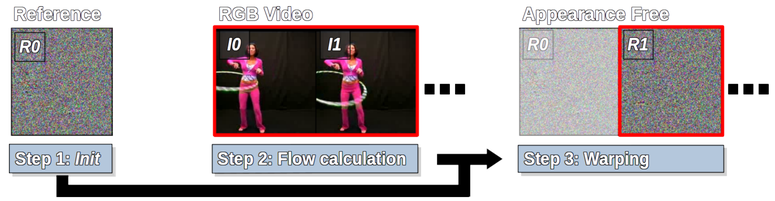
- Is my dataset or my model biased?
- Is there a way to quantify the importance of motion for particular action classes?

Pragmatically problematic

- Jurisdictions requiring explainability as a precondition for deployment of ML/AI technologies.
- Good arguments against the deployment of such systems if the relation between appearance and motion, and by extension the influence on the model is not understood clearly.

A new dataset containing only dynamics

Create a dataset with no appearance cues: AFD101



Appearance free videos are created by the following procedure

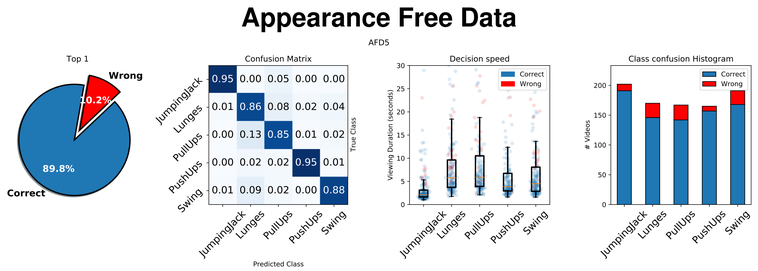
1. initializing noise *R0*
2. calculating optical flow between input frame pairs, and
3. using the optical flow to warp *R0*

Steps 2 and 3 are repeated for each frame-pair in the input video.

Psychophysical Evaluation

Userstudy with 10 participants shows:

- Humans outperform all SoTA networks on AFD.
- Decision time needed increases compared to regular video.
- Failures due to large optical flow displacement.
- Only very little variability between participant's ability.

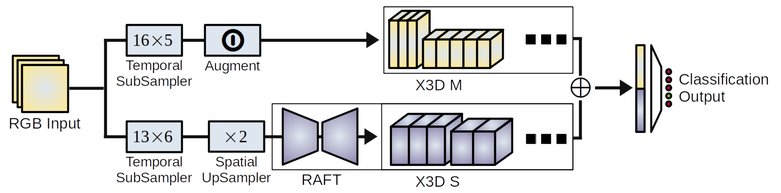


Interactive Quiz

Are you better than my colleagues?

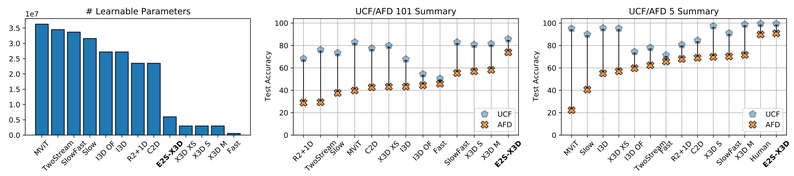
Results and Conclusions

A network architecture that works better on AFD101



Our novel ES2-X3D operates with two parallel streams for processing of RGB (top stream) and optical flow (bottom stream). Late fusion via concatenation is followed by a fully connected and Softmax layer.

Comparison with common action recognition architectures



Most of the action recognition architectures obtain around 70-80% Top-1 accuracy on regular video. On AFD data the average performance drops by approximatelly 30% almost evenly across all architectures, with a few exceptions.

Main Takeaway

> Optical flow, or a similarly descriptive representation, is not an emergent property of any of the tested action recognition architectures.

Demo & Resources

Project Page

https://f-ilic.github.io/AppearanceFreeActionRecognition